

Consent Form

Introduction

You are invited to take part in a research study about how medical students make clinical decisions and how an AI system's reasoning may influence those decisions. Before you decide whether to participate, please read the information below carefully. You may ask questions at any time by contacting the researcher.

Because this study is conducted entirely online and involves minimal risk, signing a consent form is not required. Your consent will be indicated by clicking "I agree to participate" at the bottom of this page.

Purpose of the Study

The purpose of this study is to understand how exposure to an AI agent's clinical reasoning and analysis affects decision-making and confidence in hypothetical clinical scenarios. The results may help researchers better characterize how AI systems interact with human judgment in medical contexts.

What You Will Be Asked to Do

All study activities will take place online in this Clinical Decision Annotation System.

If you choose to participate, you will complete the following steps:

Clinical Decision Annotation Process

1. Step 1 – Initial Answer: You will review a clinical question with multiple-choice options. You will select the answer you think is most appropriate and indicate your confidence level on a 0–1 scale.
2. Step 2 – After AI's Analysis: You will see the AI agent's clinical reasoning and analysis for the same question. You may then revise your answer selection and confidence level, or keep your original choice.
3. Step 3 – After Correct Answer: You will see the correct answer to the question. You may make final revisions to your answer selection and confidence level based on this information. You may also be asked a small number of additional questions about your background (for example, your stage in medical training) and your experience with the AI system. The total time to complete the study is expected to be about 45–60 minutes, depending on the number of questions you annotate.

Voluntary Participation

Your participation in this study is entirely voluntary. You may choose not to participate, you may skip any question, and you may stop at any time before submitting your responses. There is no penalty for not participating or for withdrawing early.

Risks and Discomforts

This study involves minimal risk. All questions you see are hypothetical and are used only for research purposes.

- You may experience mild psychological discomfort, stress, or self-consciousness when making clinical decisions or realizing that your answer differs from the AI system or from the correct answer.
- You may experience mild fatigue or boredom from completing multiple annotation tasks.
- There is a very small risk of loss of confidentiality inherent to any online study; this risk is reduced by the data protection measures described below.

No physical procedures, medical treatments, or real patient care are involved, and there are no clinical or academic consequences for your responses.

Benefits

There are no direct benefits to you for participating in this study. However, you may find it interesting to reflect on your own clinical reasoning and how it relates to AI-generated analysis. More broadly, your participation may help researchers better understand how AI systems influence human decisions in medical settings.

Confidentiality

Your responses will be recorded in a research database without your name or other direct personal identifiers. We will not collect information such as your full name, email address from the interface, or IP address for research data. Data will be stored on secure, password-protected systems accessible only to the research team.

When results from this study are shared in presentations or publications, they will be reported in summary form only and will not identify any individual participant. Because your data are stored without direct identifiers, it may not be possible to remove your individual data after it has been submitted.

Compensation

If you are completing this study through a platform that provides compensation (such as Prolific), you will receive the compensation amount described in the study listing. If you stop

early, you may receive prorated compensation depending on how much of the task you complete, as described in the study instructions on that platform.

Contact Information

If you have any questions about this study, you may contact:

Zirui Cheng

zirui4@illinois.edu

Prof. Yue Guo

yueg@illinois.edu

Your Consent

By selecting "I agree to participate" below, you confirm that:

- You are at least 18 years old.
- You are currently a medical student or trainee.
- You have read and understood the information on this page.
- You voluntarily agree to participate in this research study and understand that you may withdraw at any time before submitting your responses.